

Learning Objectives

1. Categorize the different **hyper**lipoproteinemia and **hypo**lipoproteinemia.
2. Explain the different biochemical pathways that could potentially be targeted pharmacologically in the management of heart diseases.
3. Identify the **statins** as the main therapeutic intervention for treating **dys**lipidemia/atherosclerosis and explain the biochemical basis of their actions.
4. Illustrate the risk factors of the metabolic syndrome and its specific lipid abnormalities.

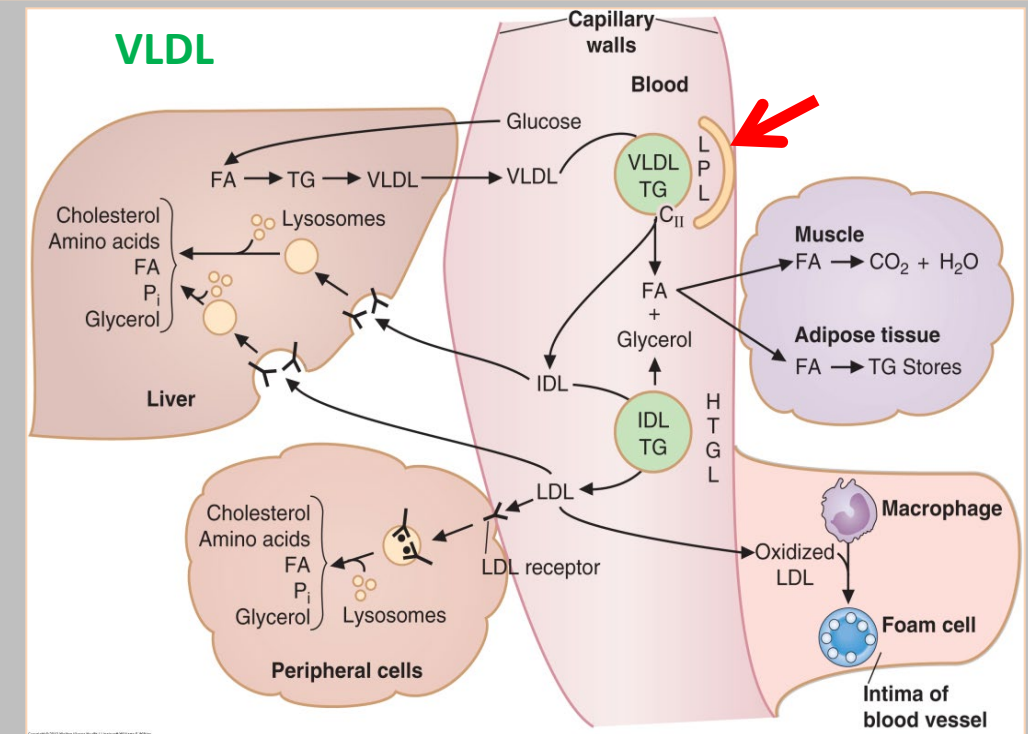
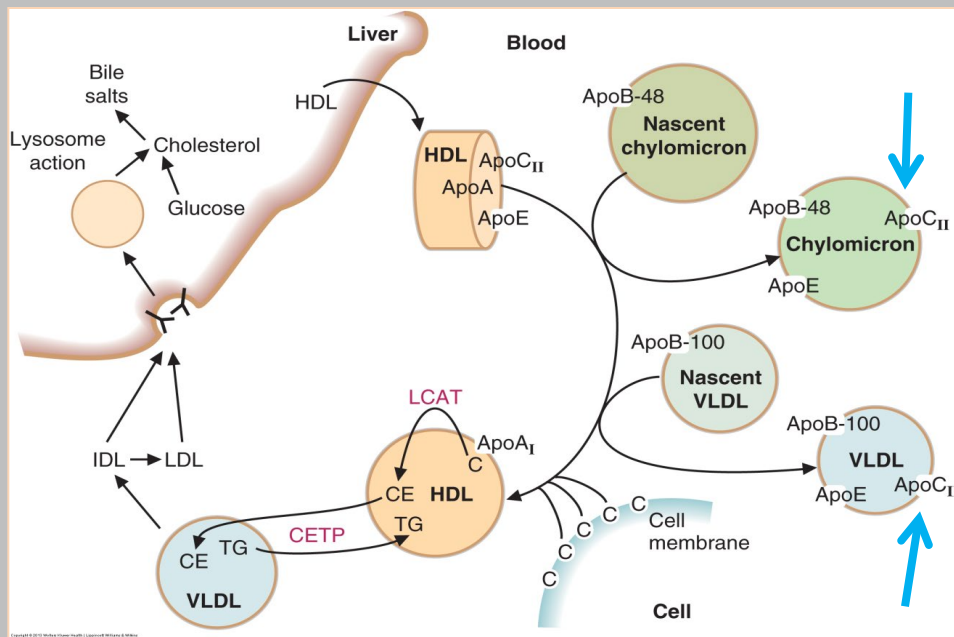
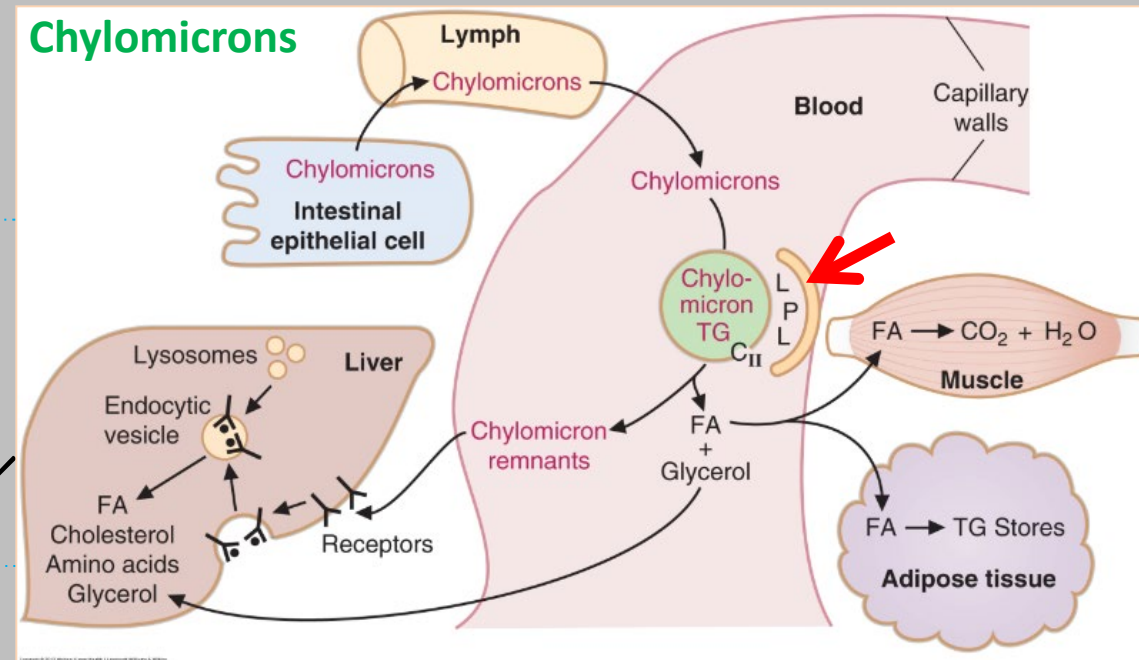
Hyperlipoproteinemia (Type I) – familial lipoprotein lipase (LPL) deficiency

Defective: **LPL** – digest triglyceride (TG)

ApoC-II – activate LPL

Consequences: chylomicron $\uparrow$, VLDL $\uparrow$, LDL $\downarrow$, HDL $\downarrow$

Increased risk of coronary disease: No



Hyperlipoproteinemia (Type IIa) – familial hypercholesterolemia

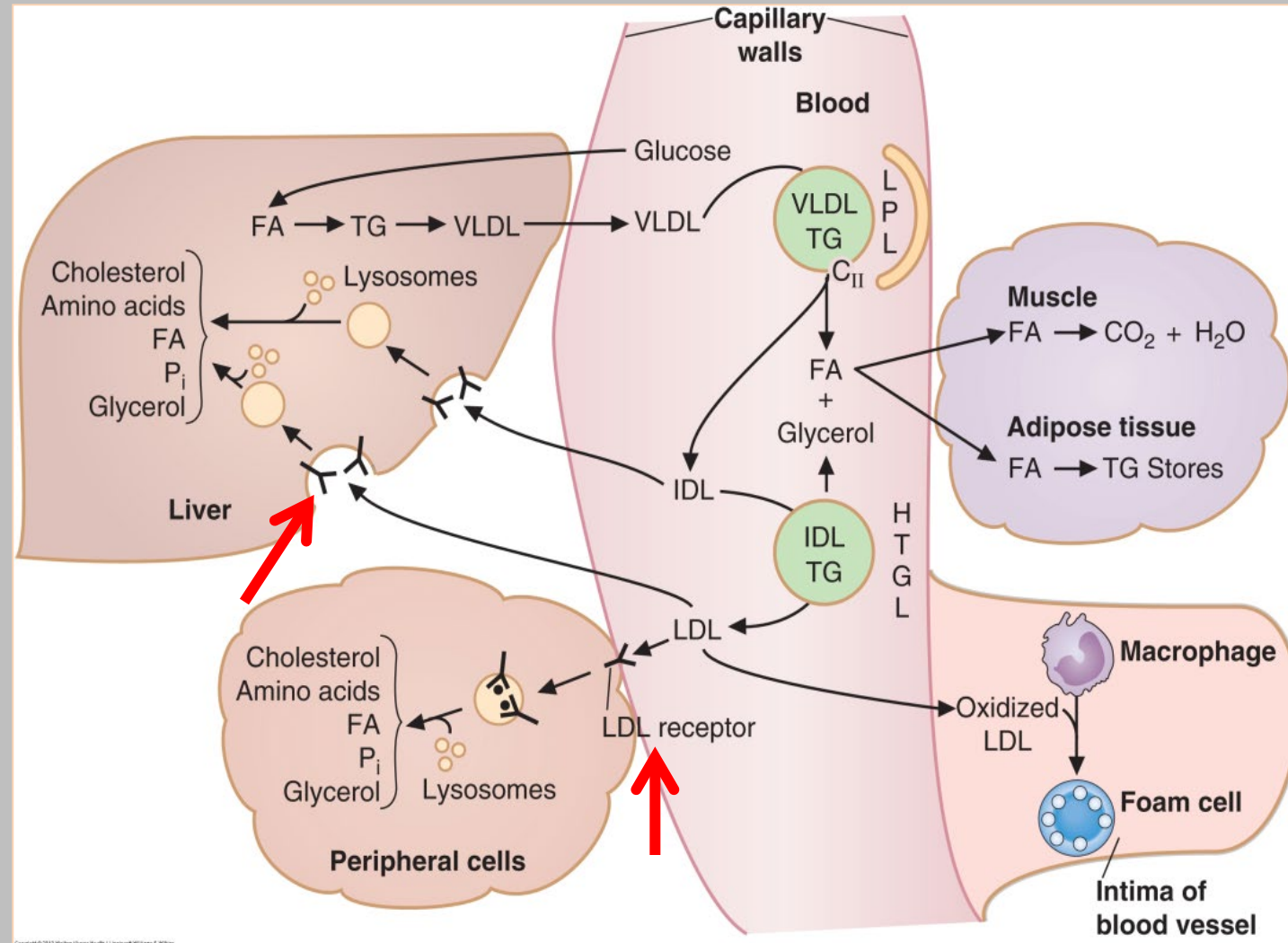
Defective: LDL receptor

ApoB-100

PCSK9

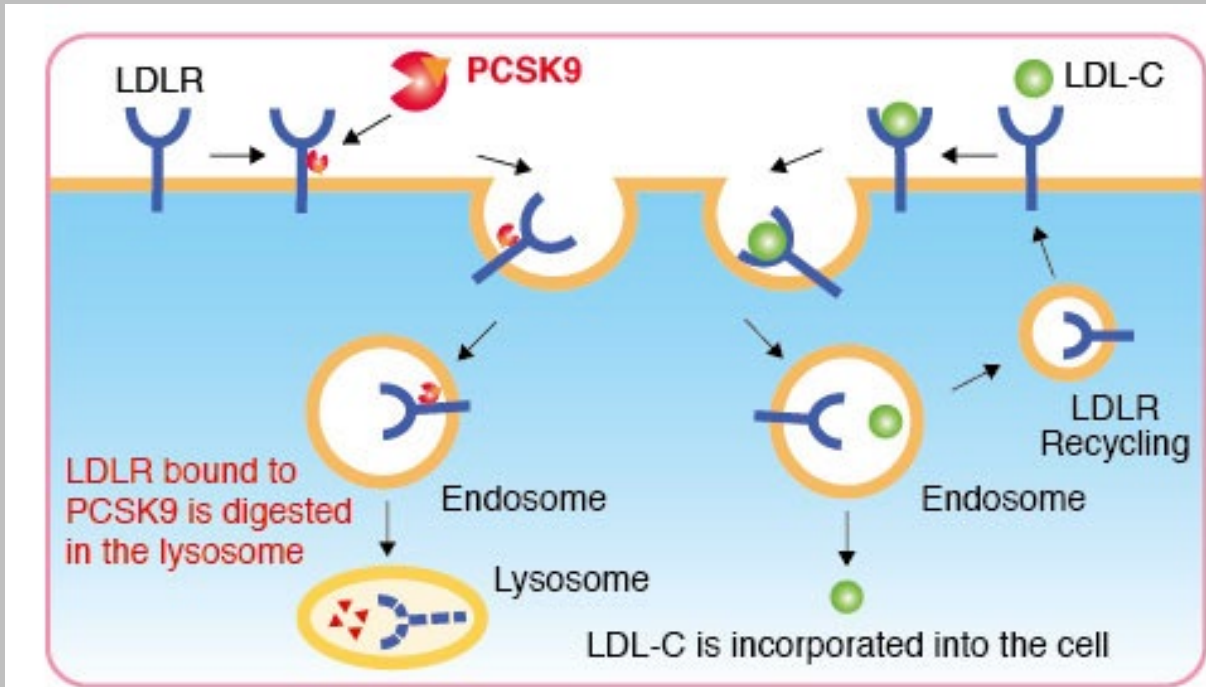
Consequences: LDL ↑, cholesterol ↑.
Increased risk of coronary disease: Yes

- **VLDL** becomes VLDL remnant or IDL after loses **TG**
- **IDL** is further metabolized to become **LDL**
- **LDL** is rich in **cholesterol**
- **LDL** is taken up by liver or peripheral tissues via **ApoB-100** binding to **LDL receptor**
- **PCSK9** regulates the turnover of LDL receptor (next slide)



Treatment of hypercholesterolemia with proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor

- PCSK9 belongs to the proprotein convertase family of serine proteases
- Secreted PCSK9 binds to LDL receptors (LDLR) and targets them for lysosomal degradation
- PCSK9 inhibition $\rightarrow$ LDLR $\uparrow$ $\rightarrow$ LDL $\downarrow$ $\rightarrow$ LDL-Cholesterol $\downarrow$
- PCSK9 inhibitors: Alirocumab (Praluent, Sanofi/Regeneron) and Evolocumab (Repatha, Amgen) are human monoclonal antibodies that inhibit the degradation of LDLR in hepatocytes and thus enhance the uptake and clearance of circulating LDL particles.



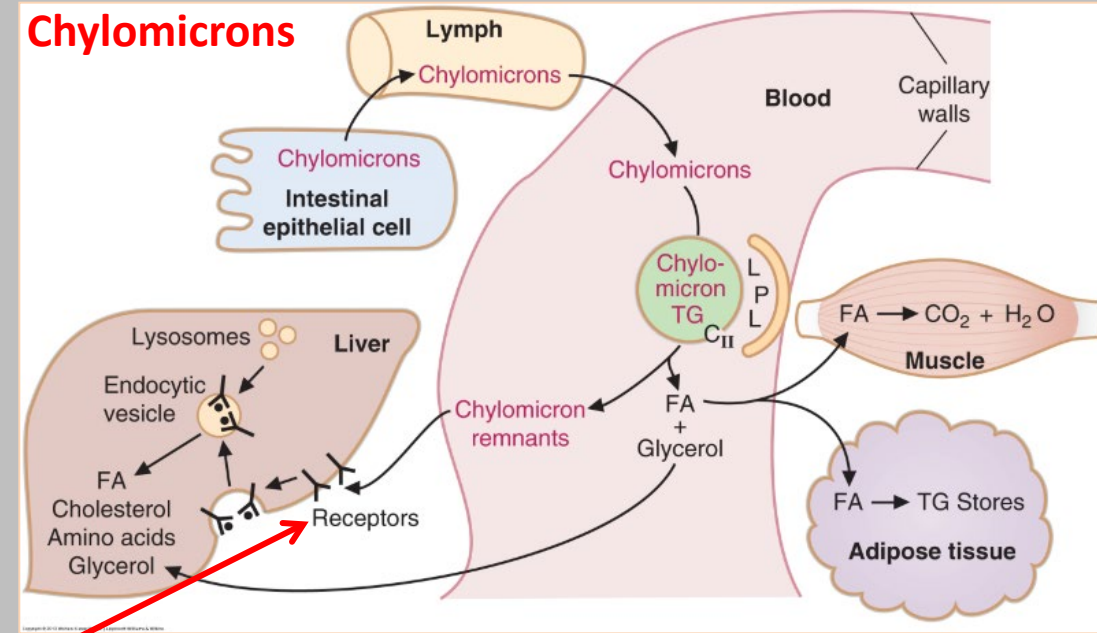
Hyperlipoproteinemia – familial Type III

Defective: ApoE

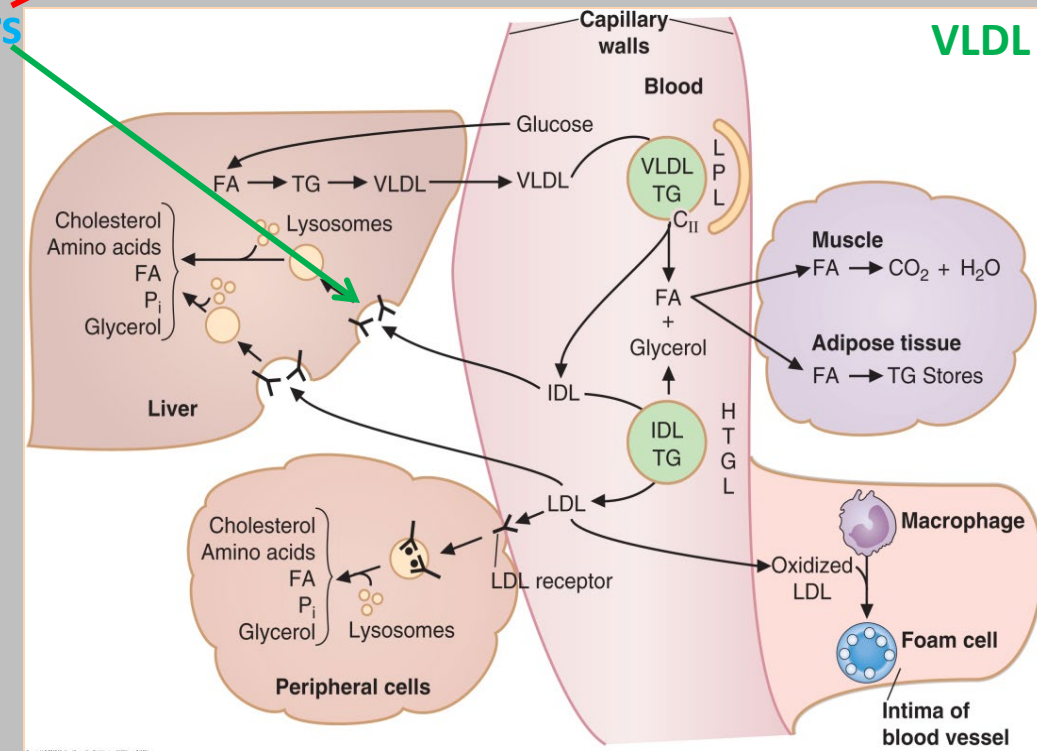
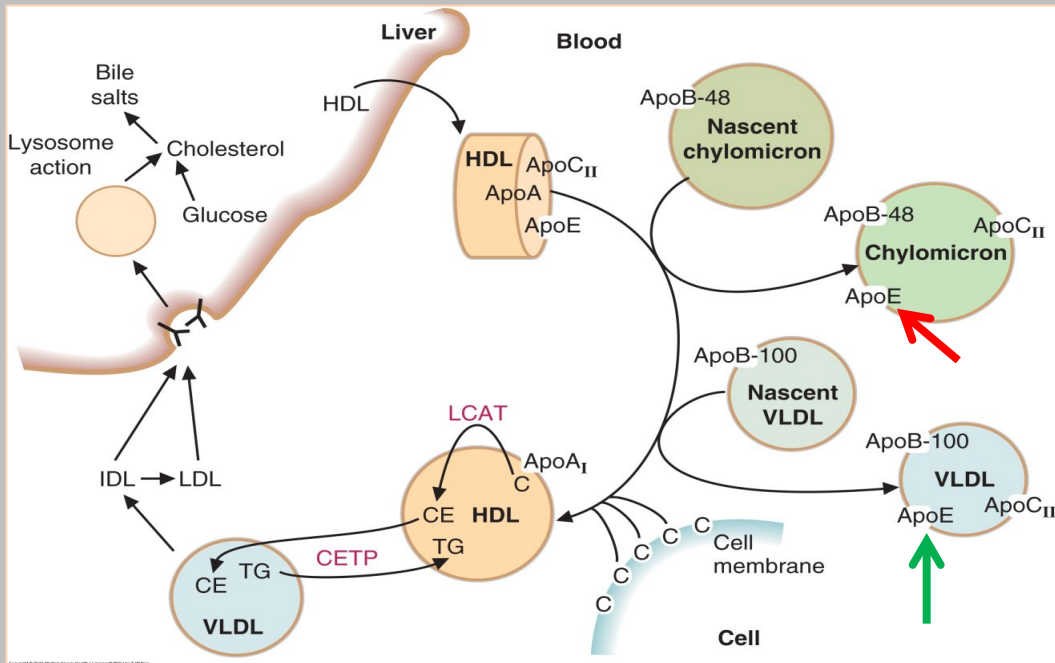
Consequences: chylomicron↑, VLDL↑

Increased risk of coronary disease: Yes

Other clinical manifestation: xanthomas



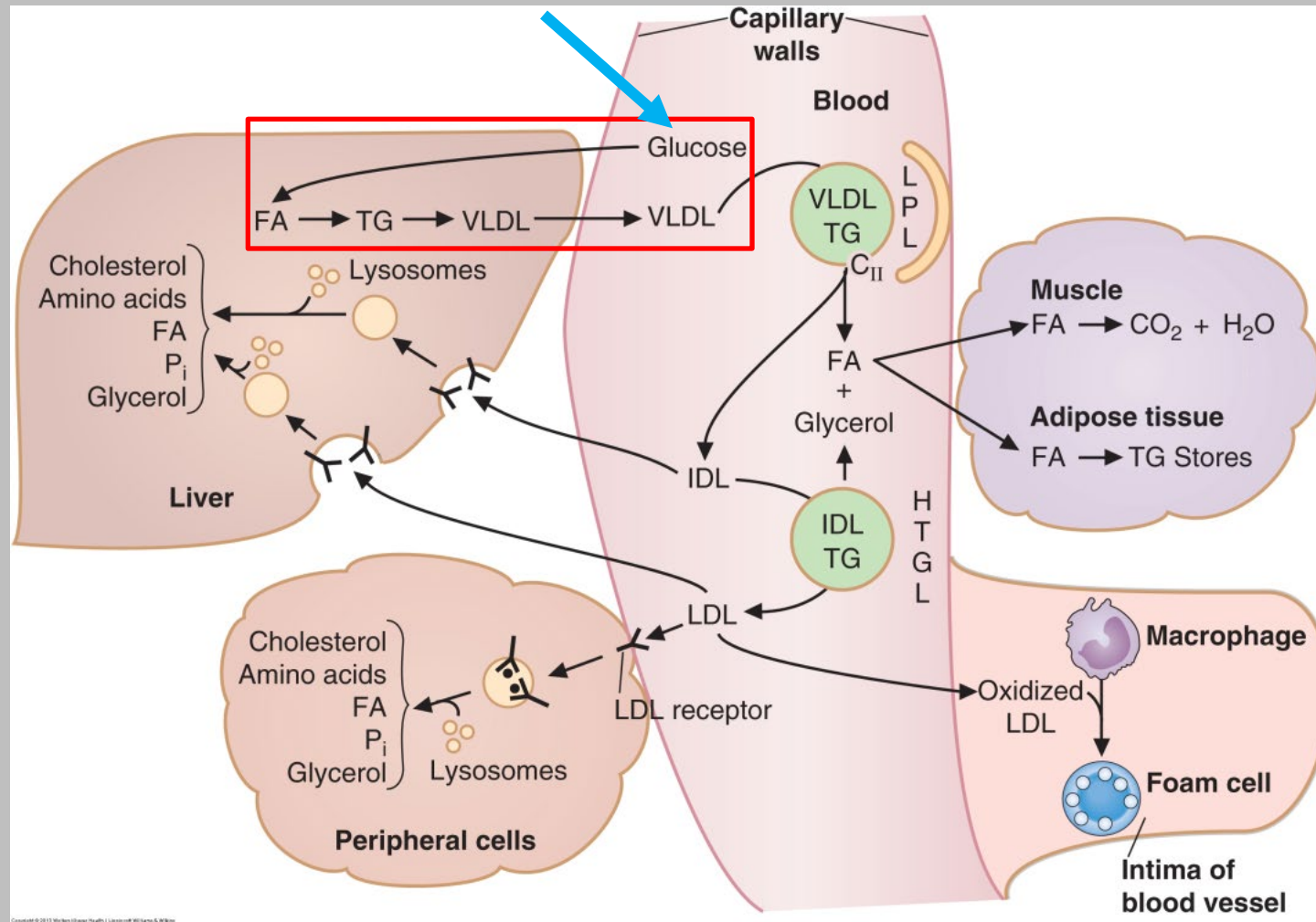
ApoE receptors



Hyperlipoproteinemia (Type IV)

– familial hypertriacylglycerolemia

Overproduction of VLDL often associated with **glucose intolerance** → VLDL ↑, TG ↑.
Commonly associated with coronary heart disease, type 2 diabetes, obesity, etc.





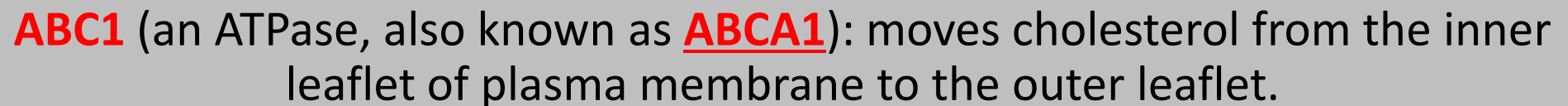
Summary of **hypolipoproteinemia**

Disorder Type	Cause	Effects
Tangier disease	Defects in ABCA1 , the transporter that supports cholesterol pickup by nascent HDLs	HDL: ↓
Abeta lipoproteinemia (Beta: Apo B -containing lipoproteins)	Mutations in the MTTP* (microsomal triglyceride transfer protein, MTP) gene	Chylomicrons: ↓ VLDL, IDL, LDL: ↓
Hypo alpha lipoproteinemia (Alpha: Apo A -containing lipoproteins)	Accelerated catabolism of ApoA-I and ApoA-II	HDL: ↓

*: The gene *MTTP* encodes the protein MTP.

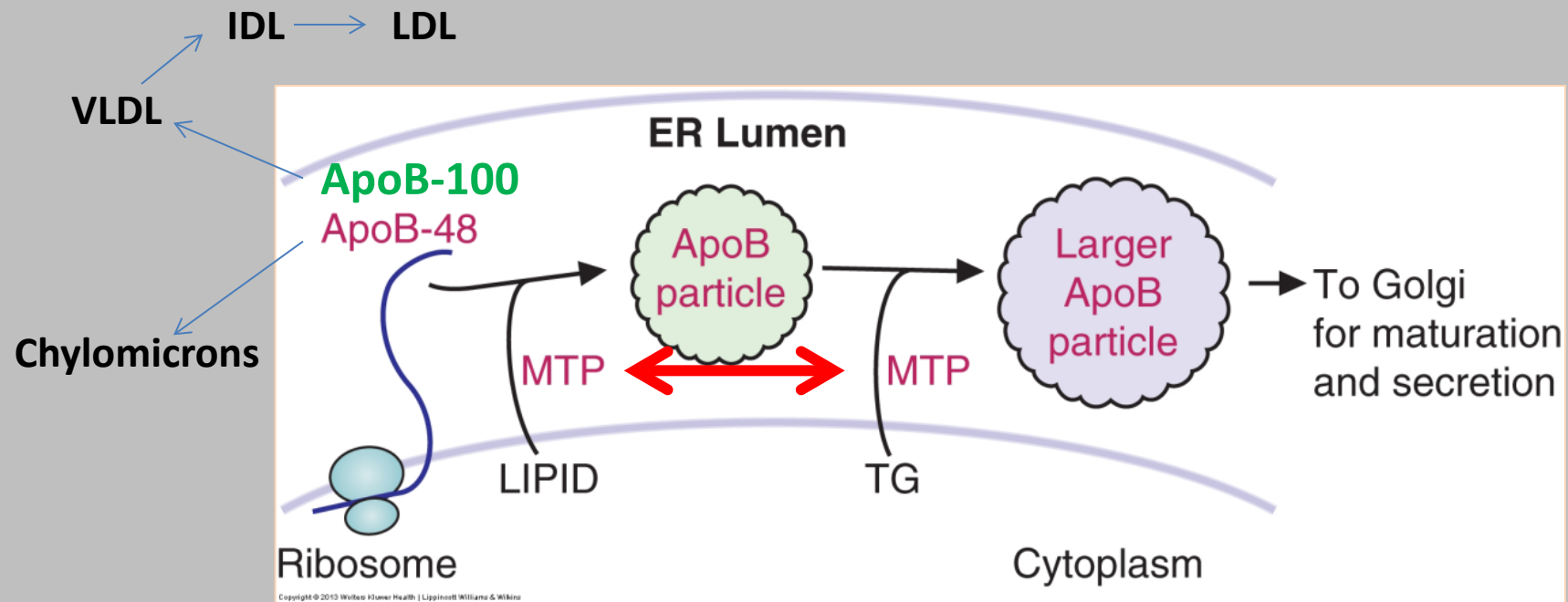
The diagram illustrates the reverse cholesterol transport pathway. It shows the liver, small intestine, and peripheral tissues. Key components include:

- LIVER**: The central organ where lipoproteins are processed.
- SMALL INTESTINE**: A red cylinder representing the site of lipoprotein absorption.
- LDL** (Low-Density Lipoprotein): Shown as a yellow sphere with a red core, moving from the liver to the small intestine.
- IDL** (Intermediate-Density Lipoprotein): A smaller yellow sphere with a red core, moving from the liver to the small intestine.
- VLDL** (Very-Low-Density Lipoprotein): A large yellow sphere with a red core, moving from the liver to the small intestine.
- HDL** (High-Density Lipoprotein): A blue sphere with a red core, moving from the small intestine to the liver.
- PC** (Phospholipids): A blue ring surrounding the red core of the HDL.
- CE** (Cholesterol Esters): A red core within the HDL.
- Apo A-I**: A blue ring surrounding the red core of the HDL.
- Apo E, C-II**: A blue ring surrounding the red core of the HDL.
- LCAT** (Lecithin:cholesterol acyltransferase): An enzyme that converts PC to lyso-PC and CE to free cholesterol.
- lyso-PC** (lysophospholipids): A blue ring surrounding the red core of the HDL.
- Free cholesterol**: A red core within the HDL.
- ABCA1** (ATP-binding cassette transporter 1): A transporter that moves free cholesterol from peripheral tissues to the HDL.
- PERIPHERAL TISSUES**: An orange area representing the source of free cholesterol.



Hypolipoproteinemia - Abetalipoproteinemia

Disorder Type	Cause	Effects
Abetalipoproteinemia (Beta: ApoB-containing lipoproteins)	Mutations in the MTTP* (microsomal triglyceride transfer protein) gene	Chylomicrons: ↓ VLDL, IDL, LDL: ↓



*: The gene *MTTP* encodes the protein MTP.

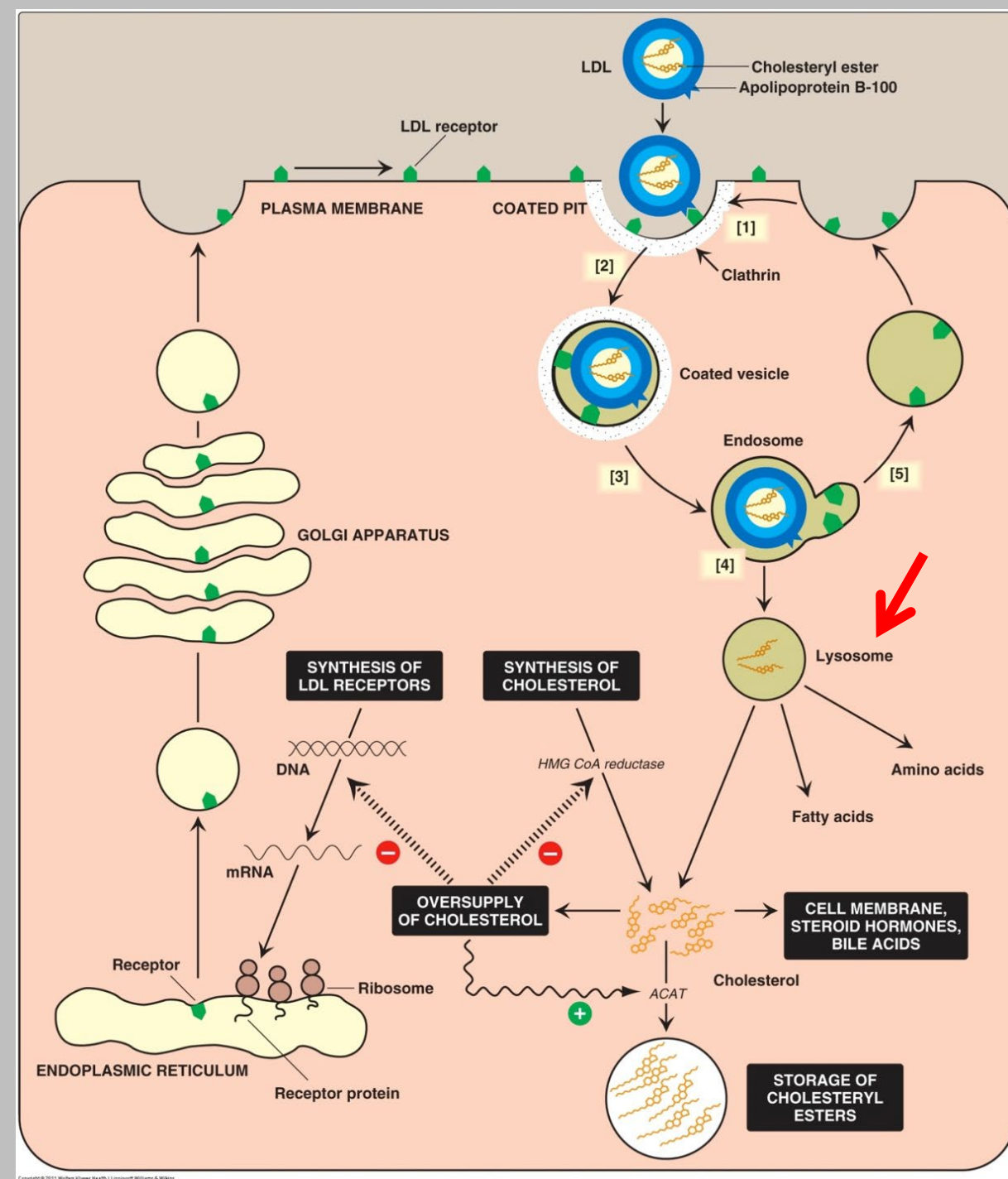
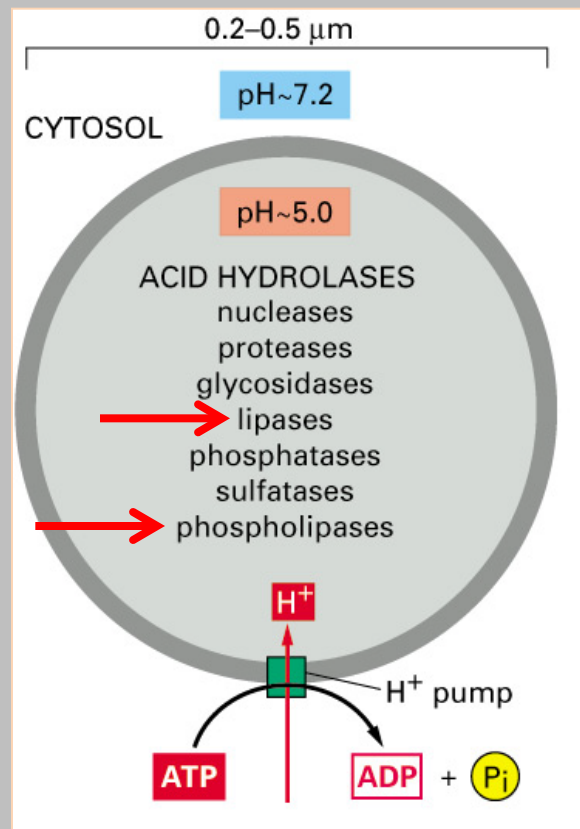


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Additional genetic diseases related to lipoproteins – **Wolman Disease**

Wolman Disease: **lysosomal acid lipases** hydrolyze **cholesterol esters** and **TGs** within the cell. If these lipases are defective, as in **Wolman disease**, the cholesterol esters and TGs accumulate in the cells and cannot be mobilized when needed, resulting in early death.

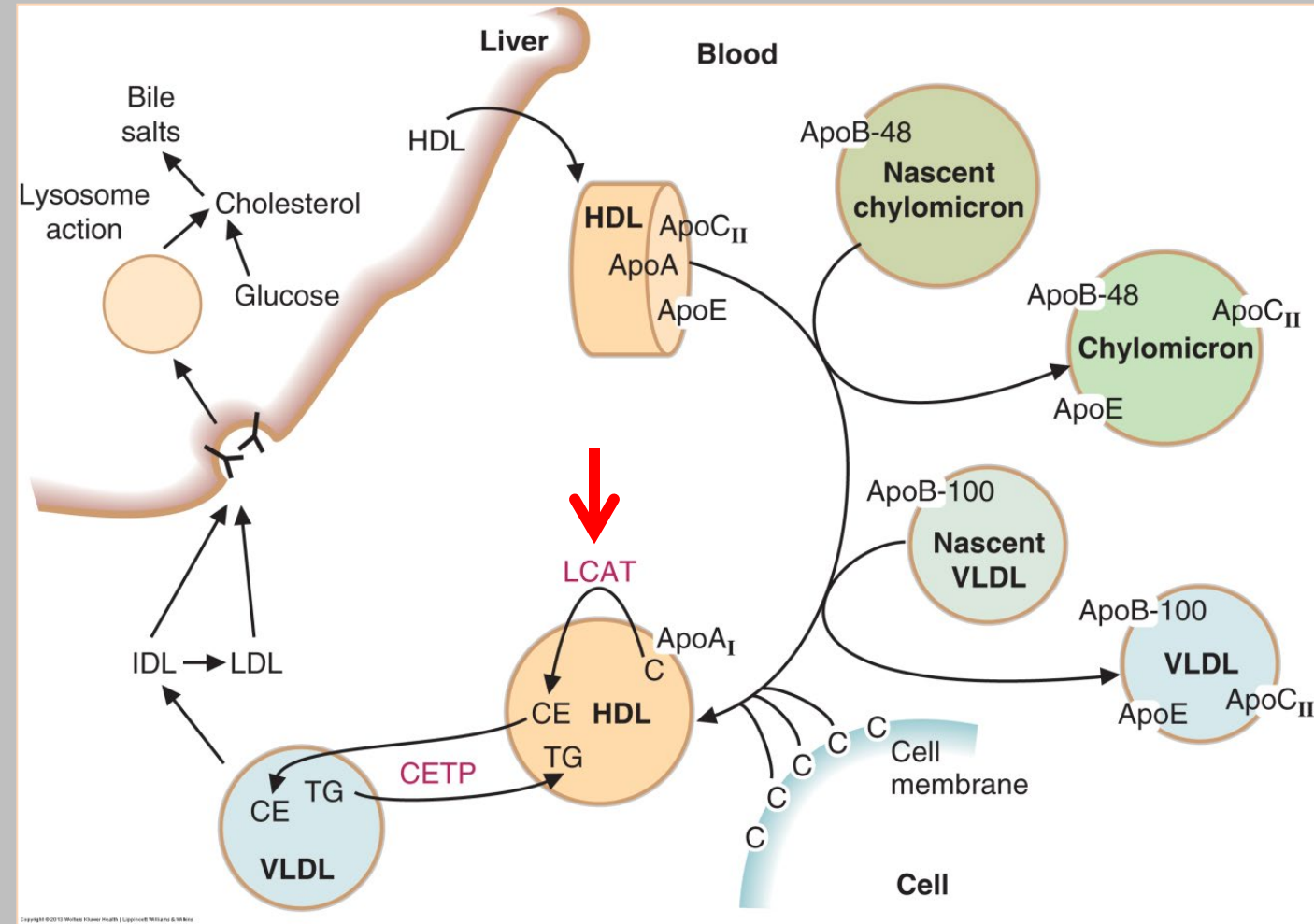




Additional genetic diseases related to lipoproteins– **LCAT deficiency**

LCAT deficiency: LCAT (lecithin - cholesterol acyl transferase) converts **cholesterol** to **cholesterol esters** in HDL.

LCAT deficiency leads to **low HDL levels** and **tissue accumulation of cholesterol**, resulting in anemia, corneal opacification, and renal failure.





Fatty Liver:

If the liver does not have adequate energy available for protein synthesis, or is starved of amino acids, or choline, this will result in **decreased synthesis of apolipoproteins or phosphatidyl choline (lecithin)**, which are necessary for the synthesis of VLDL, then the liver **cannot make VLDL**; so, most of the **TGs** that normally are exported to peripheral tissues through VLDL will not be exported, instead will accumulate in the liver and cause the **fatty liver**.



Clinical goals for cholesterol levels - HDL

- High **serum cholesterol** levels are strongly correlated with cardiovascular disease
- Higher levels of **HDL** are cardio-protective, possibly through both:
 - prevention of oxidative damage to LDL
 - reverse cholesterol transport

Control of serum cholesterol levels		HDL	
<u>Range</u>	<u>Risk</u>	<u>Range</u>	<u>Risk</u>
<200 mg/dL	low	40 mg/dL and lower	high
200 – 239 mg/dL	borderline high	60 mg/dL and higher	low
240 and up mg/dL	high		

Clinical goals for cholesterol levels - LDL

Circulating LDL can suffer **oxidative damage**; such damaged LDL are taken up by **macrophages**, leading to the formation of **foam cells**. Foam cells can accumulate in vasculatures and cause damage to them.

- LDL-cholesterol
 - Goal **<100 mg/dL** (Applies to most individuals who do not have cardiovascular disease or other risk factors for ASCVD)
 - Recommend lifestyle change at **>100 mg/dL**
 - Prescribe medications according to the ACC/AHA guidelines:
(<https://www.acc.org/~media/Non-Clinical/Files-PDFs-Excel-MS-Word-etc/Guidelines/2018/Guidelines-Made-Simple-Tool-2018-Cholesterol.pdf>)



Control of serum cholesterol levels - Drugs

- ❖ **HMG CoA reductase inhibitors** (statins: Lipitor, Zocor, Crestor etc.)
- ❖ **PCSK9 inhibitors (Alirocumab and Evolocumab)**: slow LDL receptor degradation
- ❖ **Bile acid sequestrant (Colestipol, Cholestyramine)**: increases bile acid loss in feces
- ❖ **Nicotinic acid (niacin, vitamin B3)(Niaspan, Advicor (with lovastatin))**: mechanism of action not completely understood (reduces Lp(a), as well as LDL-cholesterol and TAGs, and raises HDL)
- ❖ **Fibrate (Gemfibrizol, Clofibrate)**
 - inhibits the **hepatic release of lipoproteins** (particularly VLDL)
 - potentiates the action of **lipoprotein lipase (LPL)**
 - Increase **HDL**
- ❖ **Ezetimibe (Zetia)**
 - inhibits **dietary** absorption of cholesterol
 - combined with a statin in Vytorin



Metabolic Syndrome

The **metabolic syndrome** (syndrome X, insulin resistance syndrome): consists of a constellation of **metabolic abnormalities** that confer increased risk of cardiovascular disease (**CVD**) and **diabetes mellitus**.

Risk factors:

- A large waistline: waist circumference >102 cm (M), >88 cm (F)
- A high triglyceride level: ≥ 150 mg/dL
- Low HDL-cholesterol: <40 mg/dL, or <50 mg/dL for men and women, respectively
- High blood pressure: blood pressure ≥ 130 mmHg systolic ,or ≥ 85 mmHg diastolic
- High fasting plasma glucose level: ≥ 100 mg/dL or previously diagnosed **type 2 diabetes**

Summary

- **Key concepts:** Atherosclerosis, metabolic syndrome
- Drugs used to lower cholesterol and the biochemical basis behind them
- **Clinical correlations:** disease symptoms, biochemical basis, and treatment (Hyperlipoproteinemias, Hypolipoproteinemia, Wolman Disease, LCAT deficiency, Fatty Liver)

Q1: A 19 year old male has entered your clinic with severe chest pains, and you suspect a myocardial infarct. Blood analysis also revealed that this patient has cholesterol levels of 325 mg/dL. You tried dietary control with discouraging results. Next, you prescribe lipitor (statin) and cholestyramine. Explain the biochemical mechanisms of these drugs for lowering blood cholesterol. What vitamin in megadoses is used to lower cholesterol?